

	EPC FOR UPGRADATION OF FACILITIES AT SE FIELD			
	COMMENTS RESOLUTION SHEET (CRS)			
Contract No.	4700022908	Project No.	P30377	Comment Code - 2
Document No.		Transmittal No.	Document Title	☐ 1 Approved – No Comments ☒ 2 Approved Subject to Incorporation of Comments ☐ 3 Revise and Resubmit ☐ 4 Rejected ☐ 5 Retained for Information
P30255-14-27-76-1601	Rev A	ATC-4700022908-04458-1	FOUNDATION LAYOUT & DETAIL DRAWING FOR NEW COMPACT SUBSTATION AT WATER TANK FARM - ASAB	

S. No.	Reference / Page No.	COMPANY Comments	CONTRACTOR Response	Status / Remarks
1	SHEET 1	GRID DIMENSION AND CENTRE LINE DIMENSION OF SUPPORT IS NOT MATCHING WITH VENDOR GA DRAWING TYP. GRID LINE AT THE EDGE OF STEEL COLUMN NO AT THE CENTRELINE OF COLUMN, ALL GRIDS AND DIMENSIONS SHALL BE THOROUGHLY CHECKED WITH VENDOR DRAWING AND UPDATED FULLY.	Noted and the drawing is updated as per latest vendor GA	
2	SHEET 1	GRID NUMBER NOT MATCHING DIMENSION NOT MATCHING WITH VENDOR GA, CORRECT IT AS PER FINAL DRAWING, CHECK THE NORTH ORIENTATION WITH PSS VENDOR GA DRAWING	Noted and the drawing is updated as per latest vendor GA	
3	SHEET 1	IS ONLY THIS LIFTING SLEEVE SUFFICIENT FOR FOUNDATION LIFTING, CHECK WITH DESIGN, TRY TO PROVIDE LIFTING LUG ON FOUNDATION INSTEAD PROVIDING SLEEVE ON PEDESTAL TYP.	Noted and updated in the drawing.	
4	SHEET 1	WATER TABLE SHALL BE CHECKED WITH GEOTECH REPORT, ACCORDINGLY PROVIDE THE CONCRETE PROTECTION AGAINST GROUND WATER PROTECTION TYP.	As per Geotech report, no water table encountered.	
5	SHEET 1	LIFTING LUG TO BE DESIGN VERIFIED TYP.	Noted and presented	

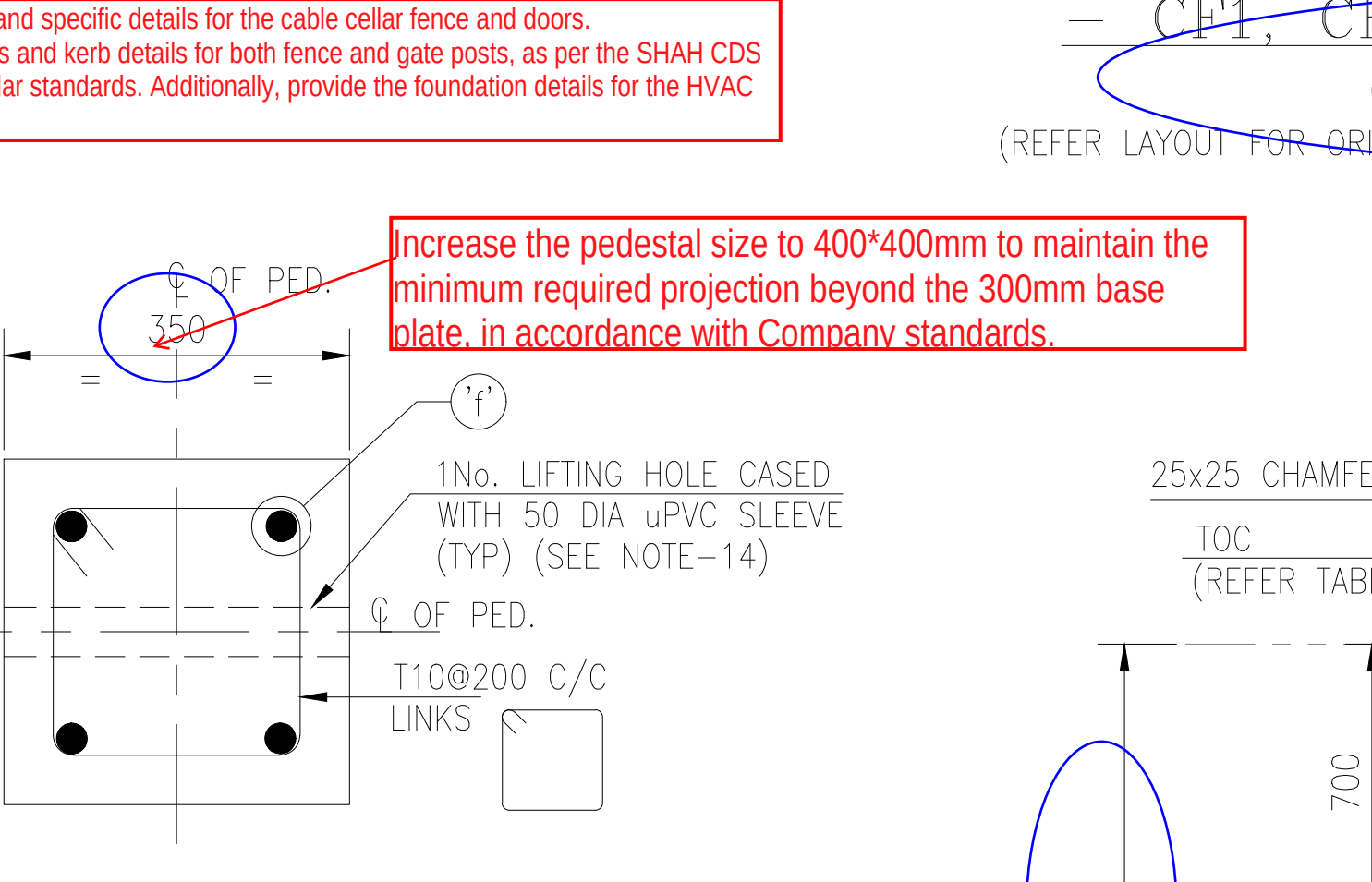
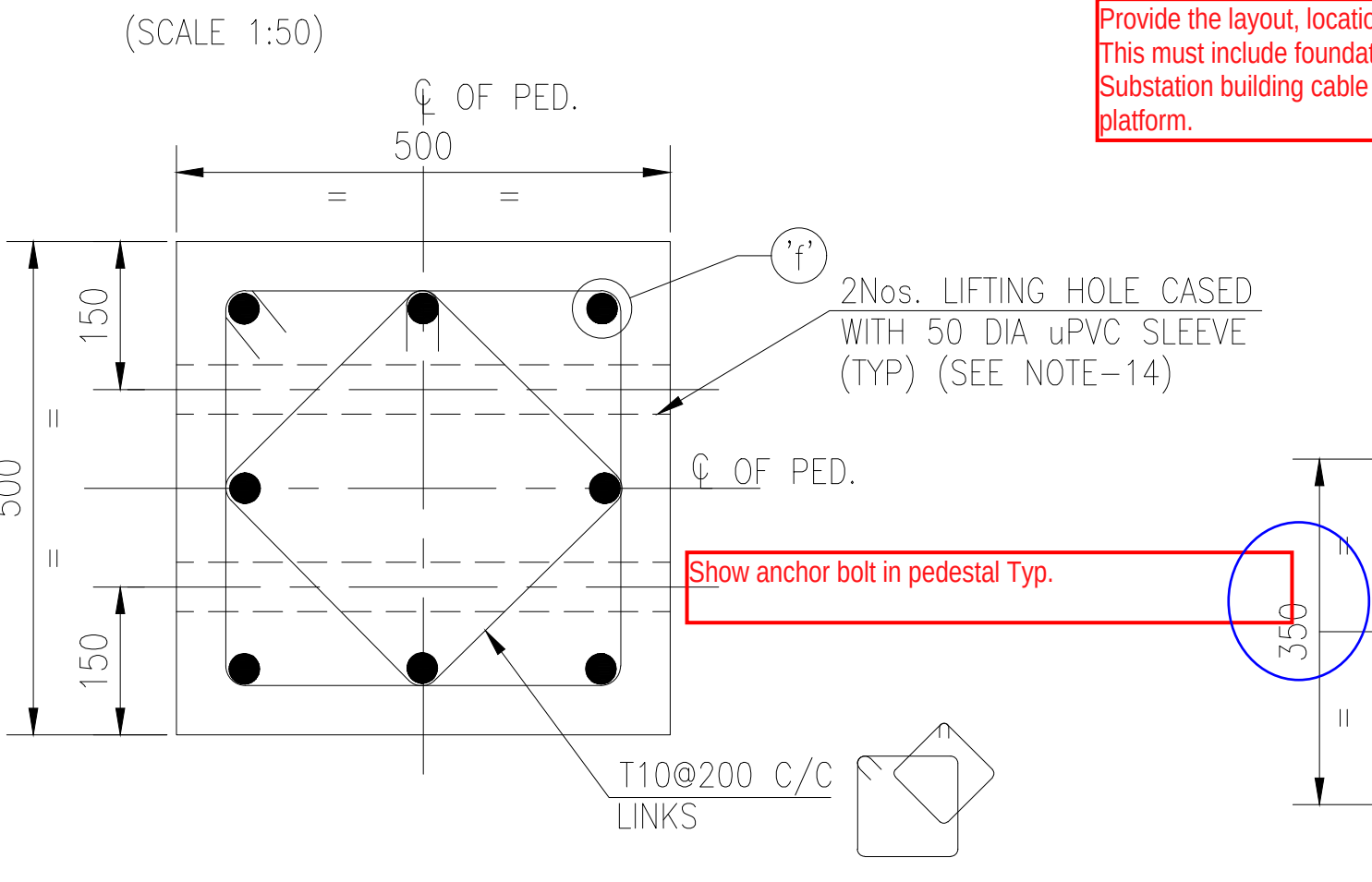
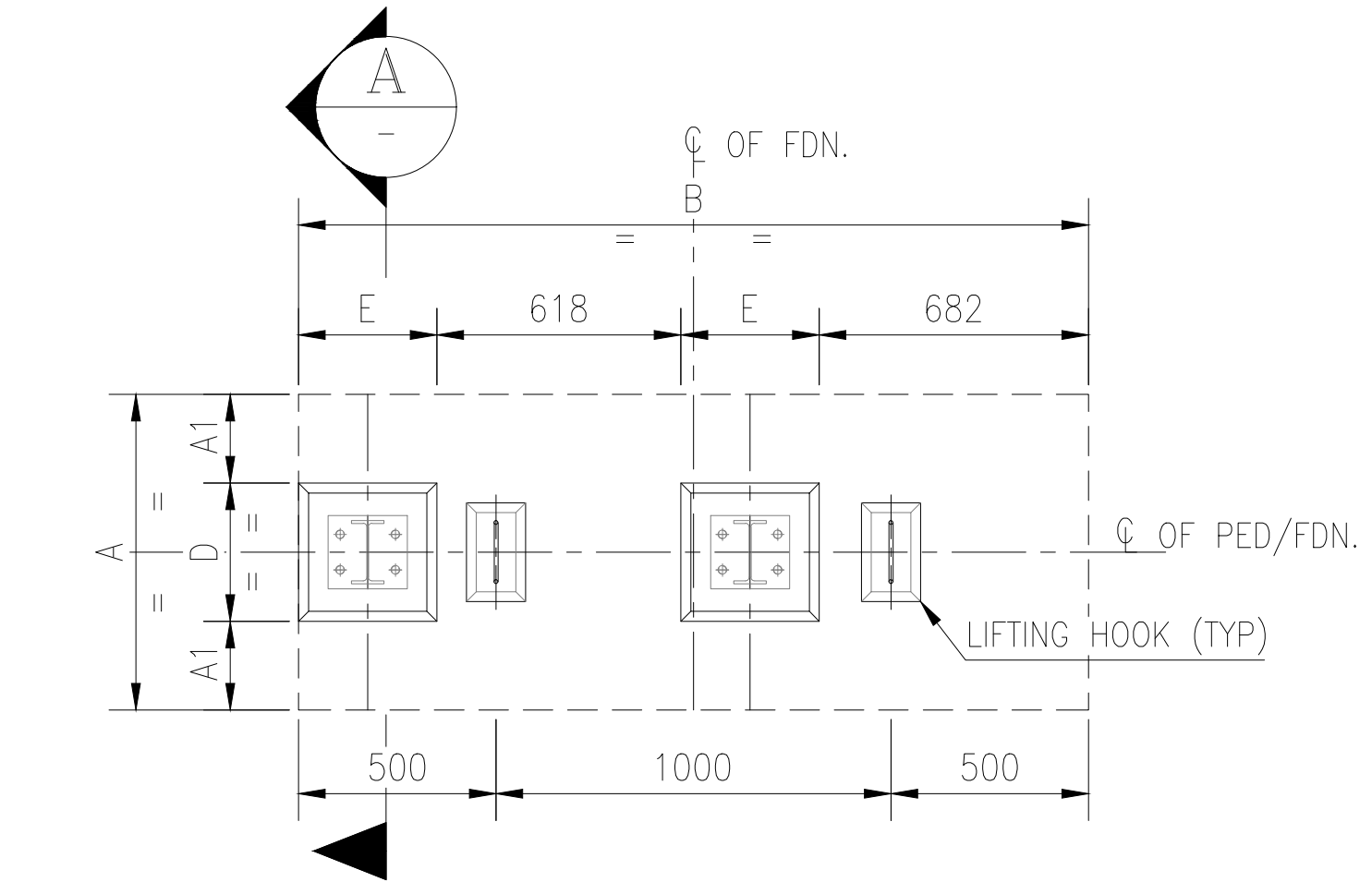
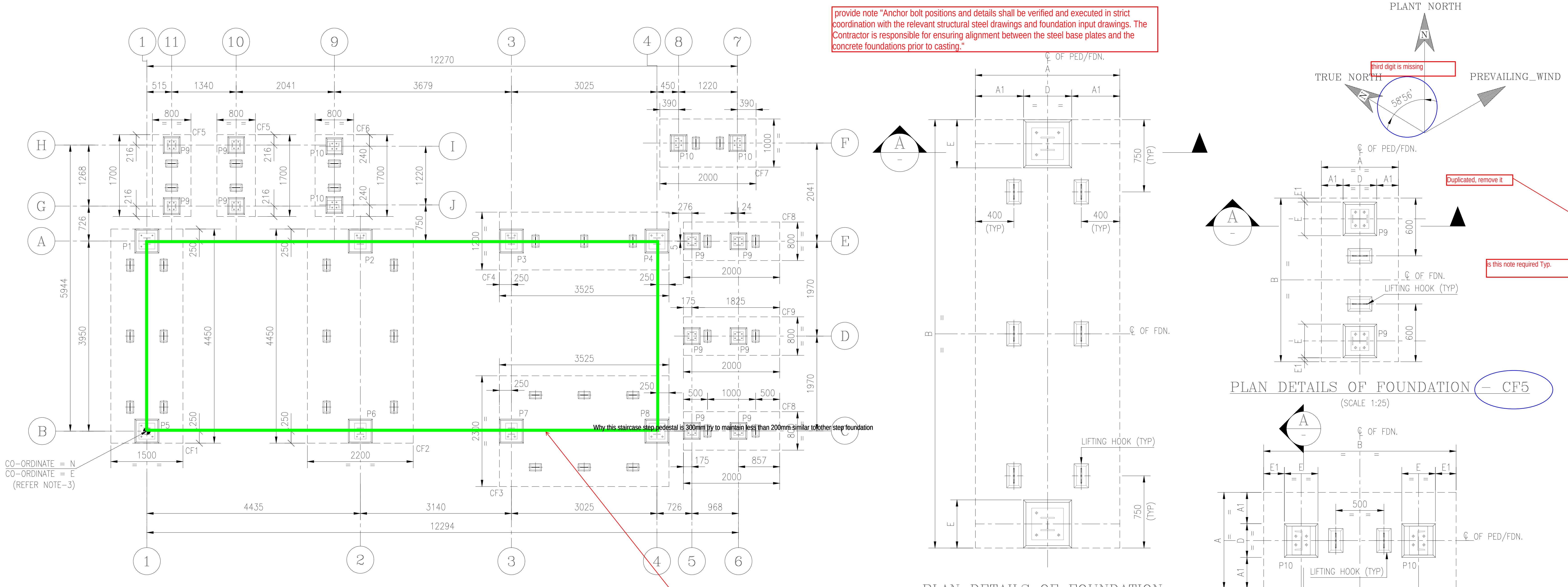
	EPC FOR UPGRADATION OF FACILITIES AT SE FIELD			
	COMMENTS RESOLUTION SHEET (CRS)			
Contract No.	4700022908	Project No.	P30377	Comment Code - 2
Document No.		Transmittal No.	Document Title	☐ 1 Approved – No Comments ☒ 2 Approved Subject to Incorporation of Comments ☐ 3 Revise and Resubmit ☐ 4 Rejected ☐ 5 Retained for Information
P30255-14-27-76-1601	Rev A	ATC-4700022908-04458-1	FOUNDATION LAYOUT & DETAIL DRAWING FOR NEW COMPACT SUBSTATION AT WATER TANK FARM - ASAB	

S. No.	Reference / Page No.	COMPANY Comments	CONTRACTOR Response	Status / Remarks
6	SHEET 1	THIS DRAWING IS PRELIMINARY SUBMISSION WITH AVAILABLE INPUTS, SAME TO BE UPDATED AS PER FINAL VENDOR INPUTS WITH PROPER CHECKING AND WITHOUT ANY DRAFTING ERRORS AND HOLD POINTS TO BE UPDATED IN NEXT REVISION WITH FINAL VENDOR INPUTS.	Noted and submitted based on the latest vendor data	
7	SHEET 1	TEXT TO BE IN SAME ALIGNMENT TYP.	Noted and updated	
8	SHEET 1	ALL THE FOUNDATION DIMENSION AND RE-BAR DETAILS, PEDESTAL HEIGHT ARE TO BE VERIFIED AND UPDATED AS PER FINAL VENDOR INPUTS AND MAT 3D DESIGN	Noted and updated	
9	SHEET 1	BOC?	Noted and corrected	
10	SHEET 1	PEDESTAL AND FOUNDATION ORIENTATION SHALL BE AS PER FOUNDATION PLAN TYP.	Noted and updated	
11	SHEET 1	CORRECT THE LEADERS OF FILLS TYP.	Noted and updated	

	EPC FOR UPGRADATION OF FACILITIES AT SE FIELD			
	COMMENTS RESOLUTION SHEET (CRS)			
Contract No.	4700022908	Project No.	P30377	Comment Code - 2
Document No.		Transmittal No.	Document Title	☐ 1 Approved – No Comments ☒ 2 Approved Subject to Incorporation of Comments ☐ 3 Revise and Resubmit ☐ 4 Rejected ☐ 5 Retained for Information
P30255-14-27-76-1601	Rev A	ATC-4700022908-04458-1	FOUNDATION LAYOUT & DETAIL DRAWING FOR NEW COMPACT SUBSTATION AT WATER TANK FARM - ASAB	

S. No.	Reference / Page No.	COMPANY Comments	CONTRACTOR Response	Status / Remarks
12	SHEET 1	CHECK THE ANCHOR BOLT SCOPE OF SUPPLY PROVIDE THE ANCHOR BOLT DETAILS, SHOW AS PER ACTUAL ANCHOR BOLT LENGTH TYP.	Noted and updated	
13	SHEET 1	NOTES SECTION SHALL BE THOROUGHLY CHECKED AND SHOW ONLY RELEVANT NOTES TYP.	Noted and updated	
14	SHEET 1	REFER DESIGN CALCULATION COMMENTS TYP.	Noted	
15	SHEET 1	PROVISIONALLY APPROVED IN CODE-2 ONLY FOR PROGRESS.	Noted	
16	SHEET 2	REFER SHEET-01 COMMENTS	Noted	
		ORIGINATOR	REVIEWED BY	APPROVED BY
Name		LAVANYA.S	KALAI SELVI	RAHUL PRAFULLA KHANDEKAR
Sign				
Date		10/12/2025	10/12/2025	10/12/2025

COPYRIGHT © 2020 ALL RIGHTS RESERVED. THIS DOCUMENT IS ISSUED IN ACCORDANCE WITH AN AGREEMENT BETWEEN ROBT STONE AND ITS CLIENTS, WHICH SETS OUT THE ENTIRE RIGHTS, OBLIGATIONS AND LIABILITIES OF THE PARTIES WITH RESPECT TO THE CONTENT AND USE OF THIS DOCUMENT.
FILE PATHWAY: C:\Users\General\Desktop\Working\30255\TRANSFORMER\COMPACT SUBSTATION TANK FARM\TO ADNOC\30255-14-27-76-1601-01.dwg



PLAN DETAILS OF FOUNDATION - CF8 & CF9 (SCALE 1:25)

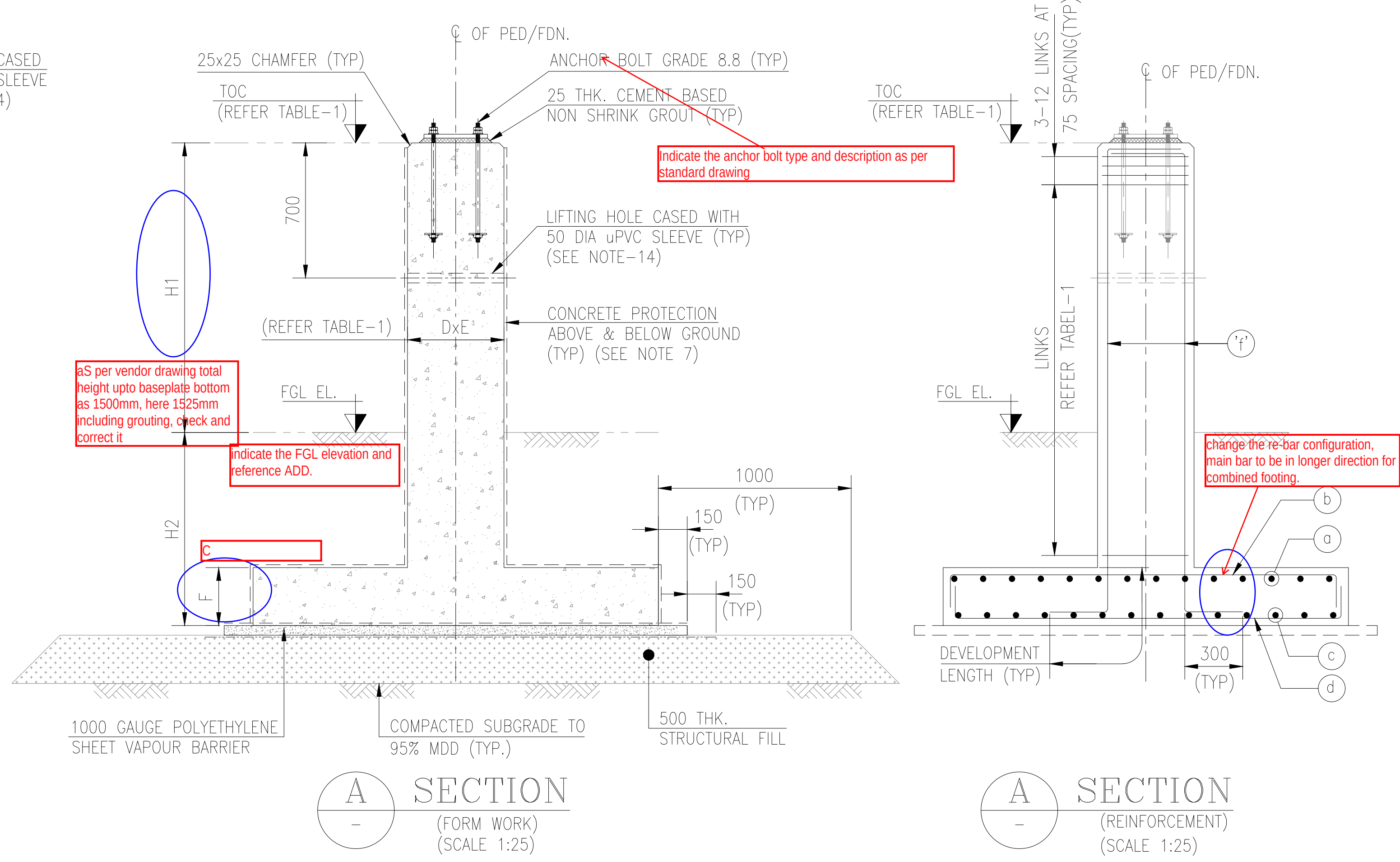
TYPICAL DETAILS OF PEDESTAL P1, P2, P3 & P4 (REINFORCEMENT) (SCALE 1:10)

TYPICAL DETAILS OF PEDESTAL P5, P6, P7 & P8 (REINFORCEMENT) (SCALE 1:10)

TABLE-1

FOUNDATION SCHEDULE																
S. NO.	FOUNDATION TYPE	FOUNDATION			PEDESTAL				TOC	FOOTING REINFORCEMENT				PEDESTAL REINFORCEMENT		REMARKS
		A	B	C	A1	D	E1	E		H2	H1	'a'	'b'	'c'	'd'	
1	CF1	1500	4450	300	500	500	–	500	1000	1500	T16@200 C/C	T16@200 C/C	T16@200 C/C	T16@200 C/C	8–16	T10@200 C/C
2	CF2	2200	4450	300	850	500	–	500	1000	1500	T16@150 C/C	T16@150 C/C	T16@150 C/C	T16@150 C/C	8–16	T10@200 C/C
3	CF3	2300	3525	300	900	500	–	500	1000	1500	T12@150 C/C	T12@150 C/C	T12@150 C/C	T12@150 C/C	8–16	T10@200 C/C
4	CF4	1200	3525	300	350	500	–	500	1000	500	T16@150 C/C	T16@150 C/C	T16@150 C/C	T16@150 C/C	8–16	T10@200 C/C
5	CF5	800	1700	300	225	350	41	350	1000	300	T12@150 C/C	T12@150 C/C	T12@150 C/C	T12@150 C/C	4–16	T10@200 C/C
6	CF6	800	1700	300	225	350	65	350	1000	300	T12@150 C/C	T12@150 C/C	T12@150 C/C	T12@150 C/C	4–16	T10@200 C/C
7	CF7	1000	2000	300	325	350	215	350	1000	300	T12@150 C/C	T12@150 C/C	T12@150 C/C	T12@150 C/C	4–16	T10@200 C/C
8	CF8	800	2000	300	225	350	–	350	1000	300	T12@200 C/C	T12@200 C/C	T12@200 C/C	T12@200 C/C	4–16	T10@200 C/C
9	CF9	800	2000	300	225	350	–	350	1000	300	T12@200 C/C	T12@200 C/C	T12@200 C/C	T12@200 C/C	4–16	T10@200 C/C

The anchor bolt configuration shall be thoroughly verified with respect to the design comments provided in the structural design calculations, ensuring alignment with the considered support conditions. This design drawing must be revised to reflect the final structural design calculations, structural drawings, and foundation input drawings (refer to Company comments). Additionally, ensure the drawing is updated in accordance with the comments provided on the foundation design calculations.



LEGEND ABBREVIATION					
CL	-	CENTER LINE	R/F	-	REINFORCEMENT
BOC	-	BOTTOM OF CONCRETE	TOC	-	TOP OF CONCRETE
EL	-	ELEVATION	TYP	-	TYPICAL
FGL	-	FINISHED GRADE LEVEL	THK	-	THICK
FDN	-	FOUNDATION		-	SET OUT POINT
PED	-	PEDESTAL			

GENERALLY REVISED



NOTES

- ALL DIMENSIONS ARE IN MILLIMETERS, CO-ORDINATES AND LEVELS ARE IN METERS UNLESS OTHERWISE NOTED.
- FOR CONCRETE & CIVIL GENERAL NOTES REFER DRAWING No. P30377-30-99-76-1626.
- FOR LOCATION, ORIENTATION, COORDINATES, ARRANGEMENT & LEVELS REFER RELEVANT CIVIL GAD'S / PLOT PLAN.
- FOR ANCHOR BOLT DETAILS REFER DRAWING No. P30377-30-99-76-1630.
- REINFORCING STEEL SHALL BE DEFORMED BARS CONFORMING TO BS 4449 GRADE 460.
- GRADE OF CONCRETE FOR RCC WORKS SHALL BE C32/40.
- FOR PROTECTION OF ABOVE AND BELOW GROUND CONCRETE SURFACE REFER DRAWING No. P30377-30-99-76-1626.
- SOIL AROUND AND BELOW FOUNDATION SHALL BE COMPACTED TO 95% MODIFIED PROCTOR DENSITY.
- THIS DRAWING SHALL BE READ IN CONJUNCTION WITH RELEVANT WELL AREA CIVIL GAD'S / PLOT PLAN AND APPROVED VENDOR DRAWINGS.
- FOR PRECAST CONSTRUCTION CONTRACTOR SHALL TAKE NECESSARY PRECAUTION TO LIFT THE PRECAST CONCRETE ELEMENTS USING THE LIFTING HOLE/LIFTING HOOK AND TRANSPORT THEM TO SITE AS PER THE APPROVED METHOD OF STATEMENT FOR TRANSPORTATION.
- MINIMUM NET SAFE BEARING CAPACITY CONSIDERED FOR DESIGN IS 160KN/m².
- UNLESS OTHERWISE SPECIFIED ALL CONCRETE EDGES OF FOUNDATIONS SHALL BE CHAMFERED 25mm.
- FOR CABLE ENTRY OPENINGS SHALL REFER RELEVANT ELECTRICAL AND INSTRUMENTATION DRAWINGS.
- LIFTING HOLES/POCKET TO BE FILLED WITH CEMENT SAND GROUT AFTER ERECTION OF PRECAST FOUNDATION IN POSITION.
- LOCATION & DETAILS OF ELECTRICAL CABLE TRENCH REFER RELEVANT WELL AREA ELECTRICAL CABLE ROUTING LAYOUT & CIVIL GAD'S DRAWINGS.
- SHALLOW LEVEL FOUNDATIONS NEAR DEEPER CABLE TRENCH SHALL BE EXECUTE AT SITE ONLY AFTER COMPLETION OF RESPECTIVE LOCATION CABLE TRENCHES.
- WEIGHT OF EACH PRECAST SHALL BE ENGRAVED ON THE PANEL IN SUCH WAY THAT IT IS CLEARLY VISIBLE.
- CLASH WITH EXISTING UNDERGROUND/ABOVE GROUND SERVICES SHALL BE VERIFIED AT SITE BEFORE EXECUTION. IN CASE OF ANY CLASH IDENTIFIED, IT SHALL BROUGHT IN TO NOTICE OF COMPANY ENGINEER AT SITE.
- CONSIDERED NET SBC 160 KN/m² SHALL BE RE-CONFIRMED AT SITE PRIOR TO THE INSTALLATION OF PRECAST FOUNDATION. IF ANY ADDITIONAL MEASURE REQUIRED TO ATTAIN THE VALUE, SAME TO BE IMPLEMENTED IN SITE AS PER AGES-SP-01-006.
- IF THE LOOSE SOIL FOUND AT THE FOUNDATION BASE THAT LOCATION FOUNDATION BASE STRATA/LOOSE SOIL SHALL BE PROPERLY REPLACED WITH WELL COMPACTED STRUCTURAL FILL TO ACHIEVE REQUIRED SBC AS PER AGES-SP-01-006.

DRAWING/DOCUMENT REFERENCES

DESCRIPTION	DWG./DOC. NO.
GENERAL NOTES FOR CONCRETE WORKS	P30377-30-99-76-1626
GENERAL NOTES FOR STEEL WORKS	P30377-30-99-76-1627
CIVIL & STRUCTURAL DESIGN BASIS	P30377-30-99-83-1601
SPECIFICATION FOR EARTHWORKS & GRADING	P30377-30-99-75-1601
TOPOGRAPHICAL AND UNDERGROUND DETECTION SURVEY REPORT	P30377-30-99-83-1604
REPORT FOR GEOTECHNICAL INVESTIGATION SURVEY	P30377-30-99-83-1605
GENERAL ARRANGEMENT DRAWING CIVIL WATER STORAGE TANK FARM	P30255-14-27-76-1601
FOUNDATION DESIGN CALCULATION FOR NEW COMPACT SUBSTATION AT WATER TANK FARM - ASAB: ACCOMMODATION	P30255-14-90-83-1604
POWER SKID STRUCTURAL GA DRAWING INCLUDING PLAN, SECTIONS AND ELEVATIONS - ASAB W/F AREA	P30255-14-27-58-3714
NAME PLATE DRAWING FOR 200KVA 11/0.433KV DISTRIBUTION TRANSFORMER 14-27-TP-4402	P30255-14-27-58-3701
ELECTRICAL EQUIPMENT LAYOUT PACKAGE SUBSTATION FOR WATER TANK FARM (INSTALLATION) ASAB	P30255-14-27-59-1609
PLOT PLAN WATER STORAGE TANK FARM ASAB (CONSTRUCTION)	P30255-14-27-04-1604
DEMOLITION LAYOUT - CIVIL ASAB: TANK FARM	P30255-14-80-79-1601
PLOT PLAN WATER STORAGE TANK FARM	MASTER-14-27-04-0010

KEY PLAN

B	10/12/2025	MMK	KS	RPK	ISSUED FOR APPROVAL
A	04/06/2025	RBA	KS/DCJ	RPK	ISSUED FOR REVIEW
REV.	DATE	DR'N.	CH'D.	AP'D.	DESCRIPTION

SCALE: AS SHOWN | LOCATION: ASAB | PROJECT No. P30255

أدنوك البرية
ADNOC Onshore

CONSULTANT / CONTRACTOR / VENDOR

ROBT STONE
ROBT STONE (ME) LLC
P.O. Box, 39588
Abu Dhabi, UAE

DRAWING No.

PROJECT: REPLACEMENT OF AGED ELECTRICAL EQUIPMENT AT SE FIELD

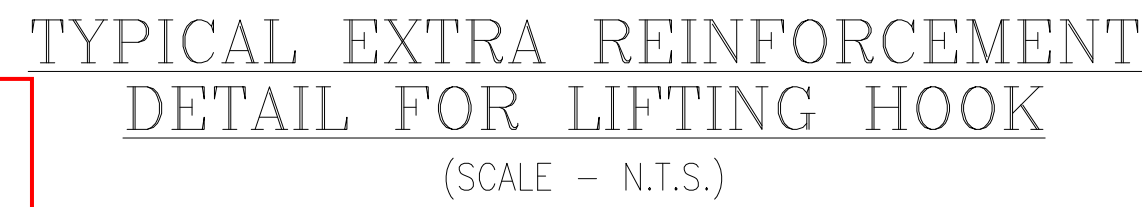
DRG. TITLE: FOUNDATION LAYOUT & DETAIL DRAWING FOR NEW COMPACT SUBSTATION AT WATER TANK FARM ASAB

DRG. No.	1427761601B	1/2
AREA	P/AREA	DOC. CODE
SERIAL No.	REV.	SH
OF	SH	SH

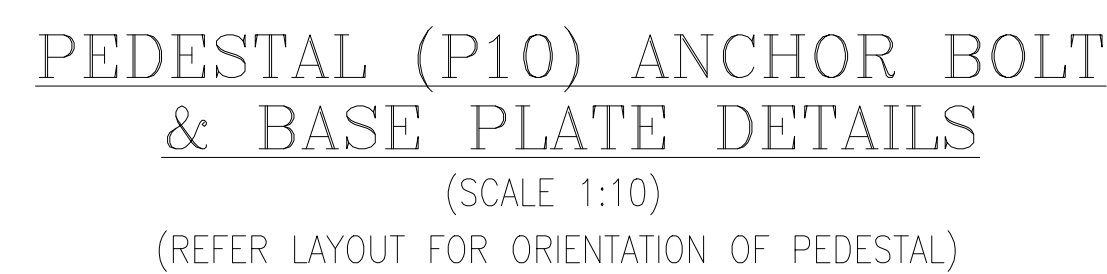
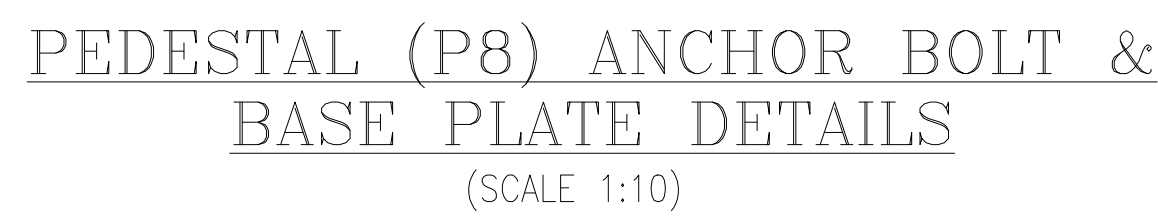
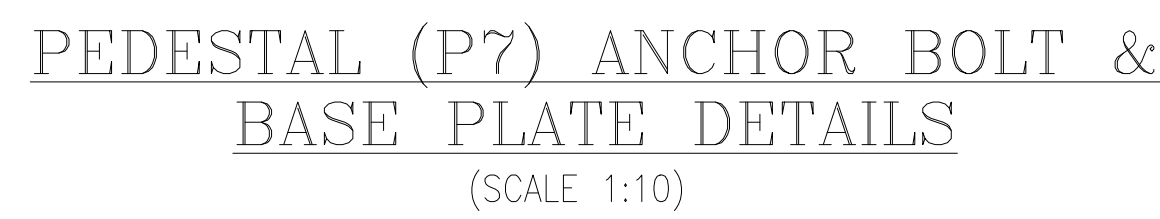
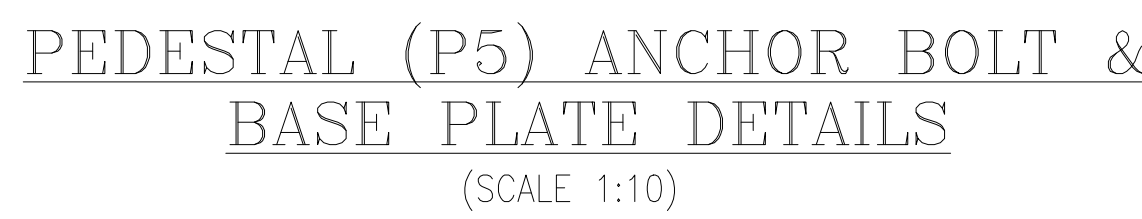
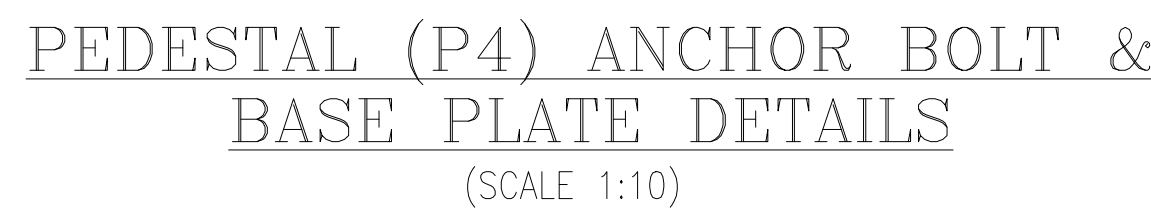
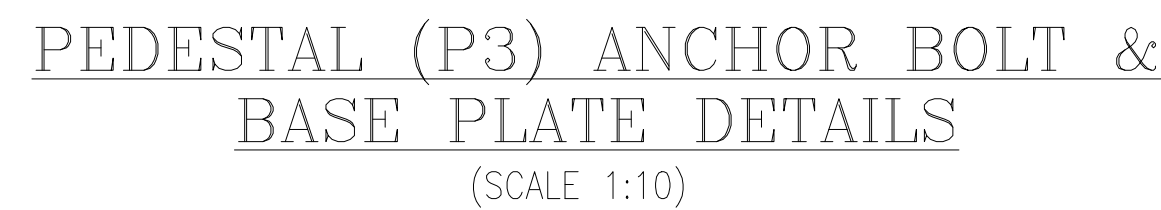
FORM_ADNOC_Onshore_A1.dwg_(REV:2)

2017 ADNOC Onshore . All rights reserved. Abu Dhabi Company for Onshore Petroleum Operations Ltd. Restricted Circulation

P30255-14-27-76-1601-01.DWG



Refer sheet-01 comments



It is the Contractor's responsibility to conduct a comprehensive check of the drawings. Any errors arising from a failure to coordinate these documents during construction will be the Contractor's liability



A triangle with the letter B inside it.



1. ALL DIMENSIONS ARE IN MILLIMETERS, CO-ORDINATES AND LEVELS ARE IN METERS UNLESS OTHERWISE NOTED.
2. FOR ALL OTHER NOTES, LEGEND AND DRAWING REFERENCES REFER SHT. 1

SCALE: AS SHOWN	LOCATION: AS AB	PROJECT No. P30255
-----------------	-----------------	--------------------



ROBT STONE (ME) LLC
P.O. Box. 39588
Abu Dhabi, U.A.E

DRG. TITLE: FOUNDATION LAYOUT & DETAIL DRAWING FOR
NEW COMPACT SUBSTATION AT WATER TANK FARM
ASAB

DRG.No.	14	27	76	160	1	B		2/2	
	AREA	P/AREA	DOC. CODE	SERIAL No.	REV.	SHT	OF	SHT	